Task 10: Industry Innovation Proposal

Proposed Solution: The Anemia Early Warning and Resource Allocation Framework (AEWRAF)

This proposal does not introduce a new analytical direction, it packages the findings of Tasks 1–9 into a single, deployable framework, because a client is better served by one coherent solution built from evidence already gathered than by a disconnected new idea. AEWRAF combines two of the brief's example categories, since our own findings point to both simultaneously: an **Early Warning System** at the clinical level, feeding a **Resource Allocation Framework** at the institutional/regional level, delivered through a shared **Decision Support Dashboard**.

10.1 Business Need

Task 1 established that this client organization is **data-rich but insight-poor**: FBC results are collected routinely across four institutions, but there is no systematic mechanism converting that data into forward-looking action. Three specific, evidenced gaps justify this proposal:

1. **No risk stratification at the point of care.** Task 5 showed that age, sex, and routine lab indices predict anemia with 94.4% accuracy, this predictive capability currently exists nowhere in the client's actual clinical workflow.
2. **No principled decision rule.** Task 8 showed that simply adopting a smarter decision threshold (rather than the conventional 0.5 default) could cut total expected error cost by 44.5% under a defensible clinical cost assumption, a substantial, currently unrealized gain achievable with no new data collection.
3. **No resource-planning visibility across institutions.** Task 4 found real, if institution-specific, prevalence differences (Institution_C at 57.3% vs. 41.6–45.5% elsewhere), and Task 1 documented systemic laboratory staffing shortages, yet there is no tool letting administrators see where risk is concentrated and plan accordingly.

10.2 Proposed Solution

AEWRAF has four components, illustrated in the conceptual framework diagram above:

1. Risk Scoring Engine. The Task 5 Elastic Net logistic regression model (age, sex, and the five non-HB red-cell indices, with the age×sex interaction) runs automatically whenever a new FBC panel is recorded, producing a calibrated probability of anemia, without requiring any new equipment or data collection, since it uses exactly the parameters already generated by standard automated hematology analyzers.

2. Bayesian Decision Layer. Rather than a fixed 0.5 cutoff, the framework applies the expected-loss threshold-selection approach demonstrated in Task 8, using a cost ratio calibrated with clinical input (our illustrative 5:1 false-negative : false-positive ratio is a starting point for that conversation, not a final answer). This layer is what turns probability into an actionable recommendation.

3. Clinical Triage Alert. For individual patients flagged above the calibrated threshold, the system generates a simple, explainable alert (not a black-box score), consistent with Task 7's finding that the original, clinically readable variables (not PCA-transformed components) should drive any deployed model, so a clinician sees "elevated risk, driven primarily by low PCV and MCHC" rather than an opaque number.

4. Resource Allocation Dashboard. At the institutional/regional level, the same underlying data aggregates into a dashboard showing anemia prevalence trends by institution, age group, and sex (extending Task 3/4's findings into an ongoing monitoring tool rather than a one-off analysis), directly supporting decisions such as where to prioritize supplementation stock or additional laboratory staffing.

A **supporting workstream**, shown as the dashed governance box in the framework diagram, runs alongside deployment rather than blocking it: validating the model across institutions using the RCBD-by-institution logic established in Task 6, and per Task 9's central recommendation - establishing consistent monthly data collection across all four institutions, which is a prerequisite for the dashboard's trend and (eventually) seasonal/forecasting components to be trustworthy.

10.3 Expected Benefits

- **Earlier identification of at-risk patients**, using data already being collected - no new tests, no new equipment, no added cost per patient.
- **A measurable reduction in costly misclassification**: Task 8's expected-loss analysis suggests a realistic path to reducing total error cost by roughly 40%, simply by replacing an arbitrary default threshold with one calibrated to actual clinical costs.
- **Evidence-based resource planning**: administrators gain visibility into where anemia burden concentrates (Task 4's institution and age×sex findings), supporting more equitable and efficient allocation of scarce laboratory and supplementation resources (Task 1's documented staffing constraints).
- **A foundation for future forecasting capability**: once Task 9's data-collection recommendation is implemented, the same dashboard infrastructure can be extended to genuine trend forecasting, following the published ARIMA precedents identified in Task 9 (the Ghana and Iraq anemia-forecasting studies).

10.4 Implementation Challenges

Being direct about these matters for a credible proposal:

- **Institutional data inconsistency** (Task 9's central finding) is the most significant blocker to the dashboard's trend-analysis component specifically, the risk-scoring and triage-alert components can deploy independently of this, since they operate on individual patient records rather than requiring a continuous time series.
- **Clinical buy-in for the cost ratio** used in the Bayesian decision layer is not a technical problem but an organizational one, it requires genuine engagement from clinical staff, not just a statistician's assumption (Task 8 flagged our illustrative 5:1 ratio as exactly that: illustrative, not empirically derived).
- **Institution-C-specific investigation** (Task 4's Test E finding) should ideally precede full multi-institution rollout, since deploying a single pooled model across institutions with a statistically confirmed, if modest, prevalence

difference risks the model performing unevenly across sites, directly motivating the Task 6-informed validation workstream.

- **Change management in a resource-constrained system.** Task 1 documented real staffing and resourcing limitations across Sri Lanka's primary care laboratory network; introducing a new clinical workflow tool, however low-cost computationally, still requires staff time to learn and trust it.

10.5 Required Resources

- **Technical:** hosting for the scoring engine and dashboard (lightweight - the underlying model is a regularized logistic regression, not a resource-intensive deep learning system); integration with existing laboratory information systems to trigger scoring automatically on new FBC results.
- **Human:** a small analytics/IT team for deployment and maintenance; clinical champions at each institution to support adoption and to inform the cost-ratio calibration (Task 8); administrative sponsorship for the data-standardization workstream (Task 9).
- **Governance:** an agreed protocol for periodically revalidating the model as new data accumulates (the feedback loop shown in the framework diagram), and clinical governance sign-off on the decision thresholds used.

10.6 Organizational Impact

AEWRAF shifts the client organization from **reactive, encounter-by-encounter interpretation** of individual FBC results (Task 1's original problem statement) toward a **systematic, evidence-based screening and planning capability**, without requiring new data collection, new equipment, or disruption to existing clinical workflows for its core risk-scoring function. It directly operationalizes this project's sub-topic (predictive screening *and* resource planning) as a single integrated system rather than two separate deliverables, and it is explicitly designed to improve over time: the Task 6 validation workstream and Task 9 data-standardization recommendation are not afterthoughts but built-in components of the framework's

own roadmap, giving the client a credible path from this one-off consultancy engagement to an ongoing, self-improving organizational capability.